

MAE 4730
Lecture 34
11/14/2018

Today:

- Rotation review
- Axis angle conversion

- compound rotations
- Q&A

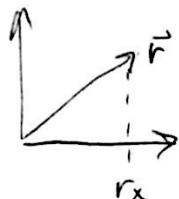
We are interested in $\vec{r}^*(t)$ fixed in β

At reference configuration: $\vec{r}^* = \vec{r}$

$$\hat{e}_i' = \hat{e}_i = \hat{b}_i$$

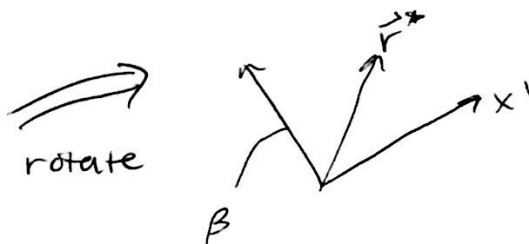
Rotation of $\vec{r} = \vec{r}^*$

↳ rotation(t)



reference config

$$\vec{r} = r_i \hat{e}_i$$



$$\vec{r}_{\beta x}^* = r_x$$

$$\vec{r}^* = r_{pi}^* \hat{e}_i'$$

$$= r_i \hat{e}_i'$$

Rotation: $\vec{r}^* = \underline{\underline{R}} \cdot \vec{r}$

rotation is linear

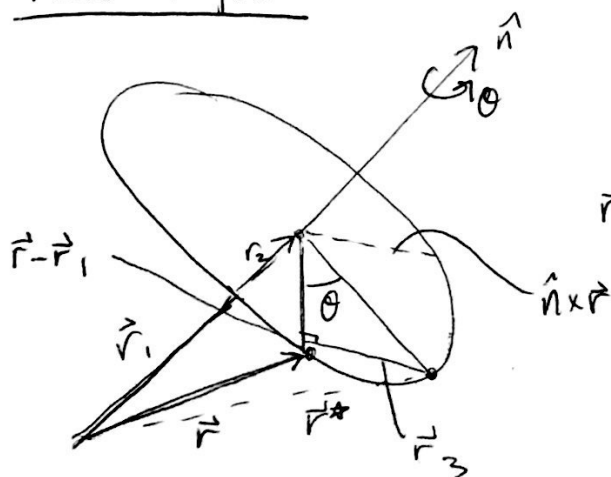
$$\underline{\underline{R}} = [R_{ij} \hat{e}_i \hat{e}_j']$$

most general tensor

$$= \hat{e}_i' \hat{e}_i$$

look at output from 3 good inputs

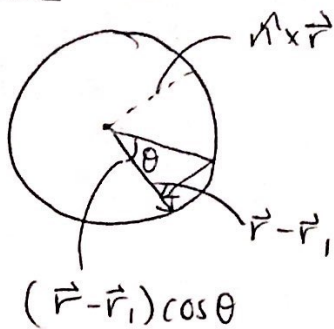
Axis - Angle:



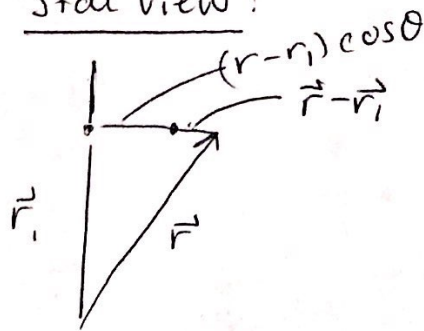
Euler's Theorem

$$\vec{r}^* = \vec{r}_1 + \vec{r}_2 + \vec{r}_3$$

Top view:



Side View:



$$\vec{r}^* = \underbrace{\vec{r}_1 + \vec{r}_2}_{\hat{n}(\hat{n} \cdot \vec{r})} + \underbrace{\vec{r}_3}_{\cos \theta (\vec{r} - \vec{r}_1)} + \sin \theta (\hat{n} \times \vec{r})$$

$$\underline{\underline{R}} = [\hat{n}\hat{n} + (\underline{\underline{1}} - \hat{n}\hat{n}) \cos \theta + \sin \theta \mathcal{L}(\hat{n})]$$

$$[\mathcal{L}(\hat{n})] = \begin{bmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{bmatrix}$$

$$[\underline{\underline{1}}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[\hat{n}\hat{n}]_{ij} = [\hat{n}]_i [\hat{n}]_j^T = \begin{bmatrix} n_1 n_1 & n_1 n_2 & n_1 n_3 \\ n_2 n_1 & n_2 n_2 & n_2 n_3 \\ n_3 n_1 & n_3 n_2 & n_3 n_3 \end{bmatrix}$$

times table

Find $\hat{n}, \theta$ given $\underline{R}$?

$$\frac{[\underline{R}] - [\underline{R}]^T}{2} = \left[\begin{array}{ccc} 0 & -n_3 \sin \theta & n_2 \sin \theta \\ & 0 & -n_1 \sin \theta \\ & & 0 \end{array} \right] \left. \vphantom{\frac{[\underline{R}] - [\underline{R}]^T}{2}} \right\} R^S$$

$$a = \begin{bmatrix} n_1 \sin \theta \\ n_2 \sin \theta \\ n_3 \sin \theta \end{bmatrix} = \begin{bmatrix} -R_{23} \\ R_{13} \\ -R_{12} \end{bmatrix}$$

$$[\hat{n}] = \frac{[a]}{|[a]|}$$

$$|[a]| = \sin \theta$$

$\Rightarrow$ know $\hat{n}, \sin \theta$

• need $\cos \theta$

$$\sum(\text{diagonal terms of } R) = 3 \cos \theta + \overbrace{(n_1 n_1 + n_2 n_2 + n_3 n_3)}^1 (1 - \cos \theta)$$

$$\text{trace}(R) = 3 \cos \theta + 1 - \cos \theta = 2 \cos \theta + 1$$

$$\cos \theta = \frac{\text{trace}(R) - 1}{2}$$

$$\boxed{\theta = \text{atan2}(\sin \theta, \cos \theta)}$$